

Challenge (Habanero)

Open Ended Questions (FRQ)

Q9. A data analyst is tracking the number of users on a gaming platform. The number of users y is related to the number of hours x by the equation $y = 5x + 20$. Find the rate of change of users per hour and explain its meaning in the context of the problem. You must show all steps and provide a clear explanation.

Q10. A programmer is optimizing the performance of a game. The game's frame rate f is related to the graphics quality q by the equation $f = \frac{120}{q}$. Find the rate of change of the frame rate with respect to the graphics quality when $q = 4$. Explain the meaning of this rate of change in the context of the problem. You must show all steps and provide a clear explanation.

Chef's Tips

- Read each question carefully
- Use the correct units
- Show all work